FONTYS ICT · DECISION THEORY · SEMESTER WEEKS 9–18

Decision Theory: curriculum and assessment

In this course you describe a decision problem as a Markov Decision Process (MDP): a model with states, actions, rewards and probabilities. A policy is a rule that says which action to take in each state. When you know the model, you can calculate a good policy from it. When you do not know the model, you can learn a good policy from data, by trying actions and seeing what happens. The methods that do this are known in AI as reinforcement learning (RL). You apply all of it to an MDP of your own.

LECTURER	FORMAT	MIDTERM
Iman Mossavat	**One half-day per week, 9 sessions**	**Week 13, small**

EXAM
Week 18, on paper

The course in one line

Your project grows next to the theory. Week 13 has a small midterm. The last two weeks are for quizzes, your project and the exam.

9	10	11	12	13	14	15	16	17	18
S1	S2	S3	S4	mid	S5	S6	S7	S8	S9

Session · Midterm · Quizzes, review, presentations · Exam

Week by week

Every session has the same shape: a short quiz on last week, a lecture block, then hands-on work on your own MDP. The homework of each week is small and feeds the next quiz.

SEMESTER WK 9
Session 1

Your first MDP

IN CLASS	Workshop: design a small MDP on paper, simulate it, present it. No solver yet.
READ	Sutton & Barto 3.1–3.3
HOMEWORK	Finish the result sheet for your MDP (uploaded once, completion only).

SEMESTER WK 10
Session 2

Returns, policies, V and Q

IN CLASS	Reading a Q-table. Spotting a flawed MDP. When is RL a sensible tool?
READ / WATCH	S&B 3.1–3.4 · video part 1
QUIZ 1	MDP design, the Markov property, return.
HOMEWORK	Write the V equations for a fixed policy on a 2–3 state slice of your MDP.

SEMESTER WK 11
Session 3

Bellman equations for a fixed policy SPRINT 1 ENDS

IN CLASS	Solve small systems by hand. Iterative policy evaluation in the course toolkit.
READ / WATCH	S&B 3.5–3.8 · video part 2 (first half)
QUIZ 2	Returns, the meaning of V and Q.
HOMEWORK	Code policy evaluation for your MDP.

SEMESTER WK 12
Session 4

Optimal policies: value iteration and policy iteration

IN CLASS	One value-iteration sweep by hand. One policy-improvement step. Generalized policy iteration.
READ / WATCH	S&B ch. 4 · video part 2 (second half)
QUIZ 3	Bellman equations for a fixed policy.
HOMEWORK	Run value iteration on your MDP and compare with the expert Q-table.

SEMESTER WK 13
Midterm

Midterm week SMALL MIDTERM

Use this week to finish your MDP solved with the model. If it does not work, show where it fails, with a clear analysis of the failure. The week also has a small midterm. `format to be confirmed`

SEMESTER WK 14
Session 5

Monte Carlo SPRINT 2 ENDS

IN CLASS	Estimate V from episodes. How many episodes do you need? Confidence intervals, back to statistics.
READ / WATCH	S&B ch. 5 · video part 3
QUIZ 4	Value iteration and policy improvement.
HOMEWORK	Monte Carlo estimate with a confidence interval for your MDP.

SEMESTER WK 15
Session 6

Exploration and off-policy learning

IN CLASS	ε-greedy exploration, learning a policy from episodes, learning about one policy while following another.
READ / WATCH	S&B ch. 5 (rest) · video part 3 (rest)
QUIZ 5	Monte Carlo and confidence intervals.
HOMEWORK	ε-greedy control on your MDP, averaged over several runs.

SEMESTER WK 16
Session 7

Temporal-difference learning and Q-learning SPRINT 3 ENDS

IN CLASS	One Q-learning update by hand. Off-policy learning. Q-learning compared with value iteration.
READ / WATCH	S&B ch. 6 · video part 4
QUIZ 6	Exploration, ε-greedy, off-policy learning.
HOMEWORK	Q-learning on your MDP. Compare its policy with the expert's.

SEMESTER WK 17
Session 8

Review, quiz and project presentations SPRINT 4 ENDS

IN CLASS	A cumulative quiz on everything so far, practice for the exam, and your project presentation.
QUIZ 7	Temporal-difference learning and Q-learning, plus a mix of earlier topics.
HOMEWORK	Prepare your cheat sheet for the exam.

SEMESTER WK 18
Session 9

Exam WRITTEN EXAM

IN CLASS	Written exam on paper, 90 minutes plus 30 extra. After the exam: last project work and wrap-up.

How you are assessed

All graded work on theory is done in class, on paper, without AI. For code you may use an AI assistant, as long as you understand every mechanism in it. I do not grade your code. I grade what you understand.

The week of the 1-1 discussion about how you built your project will be announced. `to be confirmed`

Week 18

Written exam

- On paper, in class, 90 + 30 minutes.
- Same style as the quizzes: small MDP from a story, Bellman equations, one Monte Carlo or Q-learning update by hand.
- Marks are for reasoning. A correct answer without working earns nothing.

Weeks 10–17

Quizzes

- Seven short quizzes at the start of sessions 2–8, on last week's topic.
- They are practice for the exam, in the same style, and they give you feedback on where you stand.
- One question is about your own homework or MDP.
- I mark them within the week. Going through them is part of the next session.

Weeks 9–17

Project: your own MDP

- One MDP that you keep all course. You deliver it in four sprints, each with its own target.
- It is a team project, normally with two students. Working alone is allowed.
- In a team, each student's personal contribution must be clear, and both students must have a clear technical contribution.
- You present it in week 17.
- Graded on your understanding as you show it, not on the code.

Required, not graded

Homework

- About 1–2 hours a week. Ask me for feedback whenever you want it.
- Give me access to your portfolio and submit your homework there.
- It can appear in a quiz.
- You must submit all of it to pass the semester.

Project sprints

You deliver the project in four sprints. Each sprint has one target. In a team, show in every sprint what each student contributed.

SPRINT	ENDS	TARGET
1	Week 11	**Your MDP, tested.** Idea and story agreed with me. Your MDP code tested on a tiny MDP (2 states, 2 actions) whose expected behaviour you can argue for.
2	Week 14	**Solved with the model.** You explain your model and code. Policy evaluation and value iteration on your MDP, checked against the expert Q-table.
3	Week 16	**Learned without the model.** Monte Carlo and Q-learning on your MDP, compared with the answer from the model. A minimum viable answer to every item.
4	Week 17	**Final delivery and presentation.** What you changed after the feedback, your conclusions, and your presentation.

To pass

- A grade of 5.5 or higher on the exam and on the midterm.
- An acceptable end project, delivered gradually in sprints.
- A 1-1 discussion with me about how you built your project.
- All homework submitted.

Materials

Book

Sutton & Barto, *Reinforcement Learning: An Introduction*, 2nd edition. Free from the authors.

Download the PDF

Slides and course files

The slides of the course, a workshop reader and a Python notebook. The slides are the 2023–24 version and will be updated during the course.

Open the course folder

Video series

These videos are not by Sutton and Barto. They are by the YouTube channel **Mutual Information**, in the series *Reinforcement Learning, by the Book*. They follow the chapters of the book. Watch parts 1–4. Part 5 is not part of this course.

- Part 1 · S&B 3.1–3.4
- Part 2 · S&B 3.5–3.8, ch. 4
- Part 3 · S&B ch. 5
- Part 4 · S&B ch. 6

On Canvas

The course toolkit (Python with numpy and matplotlib) and the MDP idea sheets. You need a laptop with Python.

Draft for discussion. Weeks can still change.